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CUDA Constant and Shared Memory Quiz

Graded Quiz • 15 min

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Nvidia Shared/Constant Memory Hardware

Nvidia GPU Shared and Constant Memory Analysis

Nvidia Shared Memory Allocation

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Video: GPU Device Constant Memory Allocation

4 min

Quiz: CUDA Constant and Shared Memory Quiz

Submitted

Video: CUDA Shared and Constant Memory Image Processing Lab Overview Video

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Lab: CUDA Shared and Constant Memory Image Processing Lab

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Video: CUDA Shared and Constant Memory Image Manipulation Assignment Overview Video

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Discussion Prompt: Shared and Constant Memory Discussion

10 min

🎉 Congratulations! You passed!

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CUDA Constant and Shared Memory Quiz

Review Learning Objectives

1. Constant memory size on a GPU is in direct relation to the number of streaming multiprocessors (SMs). 1 / 1 point

☐ True

☒ False

COOCH

Ready to review what you've learned before taking the quiz? I'm here to help.

👍 Correct

This is the correct answer. Based on the GPU device, there is a fixed maximum amount of constant memory that can be allocated, which is not based on the number of SMs.

🎯 Submit your assignment

Due Sep 2, 11:59 PM IST

Try again

☐ True

☒ False

Receive grade

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☐ Correct

This is true because shared memory is fairly small, below 200KBs per SM and started at 16 KB per SM with at most dozens of SMs on the most powerful devices.

👍 Like

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2. Shared memory sizes for GPU devices has always been larger than 128 MB. 1 / 1 point

☐ True

☒ False

Correct

This is true because shared memory is fairly small, below 200KBs per SM and started at 16 KB per SM with at most dozens of SMs on the most powerful devices.

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3. The extern keyword is used in conjunction with shared memory allocation in which execution target(s) to indicate dynamic memory will be shared on the GPU card? 1 / 1 point

☒ global

☐ Correct

This keyword indicates that the function will executed on the device, and in this case shared memory would be dynamic.

☒ device

☐ Correct

This keyword indicates that this function is executed on the GPU and being called by a function also on the device. When the extern keyword is used during memory allocation on the GPU, the memory available is dynamic, determined during runtime.

☐ async

☐ host

4. Which keyword is used as part of shared memory allocation? 1 / 1 point

☐ __constant__

☐ shared

☒ __shared__

☐ func

Correct

This is the correct answer.

5. When writing code that uses shared memory, which of the following are valid considerations (pros and cons) for whether to use this type of memory? 1 / 1 point

☐ Shared memory is always faster than register memory.

☒ The value in shared memory may change based on the logical and physical order of reads and writes, so tools such as thread barriers will need to be considered.

☐ Correct

This is correct, as you will need to consider this when developing device kernels.

☒ Shared memory can be faster than global memory

☐ Correct

This is correct because if the size of shared memory allocation is within per SM limits, it will be faster than global memory.

☐ Shared memory is really large and you will never over allocate this type of memory and therefore never have read-misses.

☒ Shared memory is an mechanism for threads within the same block to share data in a fast manner.

☐ Correct

This is correct. It is a common way for threads within a block to communicate, such as giving access to interim results that will not be placed in global memory.

6. Which function is used to allocate transfer data from the host to device constant memory? 1 / 1 point

☐ cudaMalloc

☐ cudaMemcpy

☒ cudaMemcpyToSymbol

☐ __constant__

Correct

This is the correct function, as its sole use is in copying data from host to device constant memory.

https://www.coursera.org/learn/introduction-to-parallel-programming-with-cuda/exam/D82yX/cuda-constant-and-shared-memory-quiz/attempt/hedirectToCover=true

1/1